

Sealed Experiment Record

Sealed 2026-10-07 · Signed by the K-Veritas server
Session 192b2a6a · Data hash 4524...4562

Abstract

This record seals 1 run (kv_run.sh) executed on Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz between 2026-10-07 17:51 and 19:55 UTC. Reported results: fid_50k 3.64607. 1 invocation in total, 0 failed. Produced with K-Veritas [1].

What this record proves

These numbers came from the sealed code at this time, unchanged since. Whether the experiment is sound is for the reader to judge.

Verify: kveritas verify <this file> or kveritas.org/verify

1 Setup

Item	Value
Session	192b2a6a-8cdd-470c-84b2-07d5e0fc198a
Initialized	2026-10-07T17:51:47Z
Machine	167ec0d6dc9a8170
Operating system	linux / amd64
CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
Memory	31.0 GB
GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
Environment, run 1	f4ac160cb6f4515783ae8044f006fd8f745f4327d164aacc938e05e08247b815
Disclosure	open
Source code, run 1	7d30756f11d3748e1247a041a72fda20c0a8f7294fd41dd2ad4aa6b13b8be119

2 Runs

#	Command	Duration	Exit	Anchored
1	bash kv_run.sh	2 hours 3 minutes	0	at run end

1 invocation in total: 1 kept in this record, 0 failed, 0 interrupted. Every invocation is recorded by the server, including the ones not kept.

3 Results

Run 1: kv_run.sh

Metric	Final value	Step	Source
fid_50k	3.64607		explicit

Inline claim	Value	Phase	Output line
fid_50k	3.64607		61

4 Execution evidence

Coherence

Coherent (0.58): the telemetry channels move together, as one process.

Telemetry

Run	Samples	CPU time	Peak memory	Peak GPU
1	7200	7524.7 s	2.67 GB	99%

5 Provenance

Run 1: kv_run.sh

Disclosure open. 169 tracked files, 3 snapshots. Every snapshot is bound into the signature.

#	Snapshot	Time	Root	Changed
0	run_start	17:51:48	1dc3d3e61639382b	169
2	run_end	19:55:44	1dc3d3e61639382b	0

1 snapshot with no file change omitted here; full timeline in Appendix B.

Activity: 189 files read, 0 files written, 2 subprocesses. Full list in Appendix B.

6 Verification

Field	Value
Signer	Signed by the K-Veritas server
Signed at	2026-10-07T19:55:47.551320+00:00
Algorithm	RSA-PSS-SHA256, 4096-bit key
Key fingerprint	eeac81653cf44ac029c4069cf2a06d9f
Data hash	45245b62b22d14bef244aff320dcb53ce6466f333abdaa5ae41a55bd7e184562
Code bundle hash	8d2399cb0fe250bae6ec09526af5f450e2875a02ce648669f0657b371dbe60f1

Run anchors: 1 invocations were anchored at the server when they ended.

Checks a verifier runs

1. The stored canonical data hashes to the data hash above, and the signature verifies over data_hash:nonce:signed_at.
2. The signing key is the K-Veritas server key, not one supplied by the author.
3. Nothing was appended after the seal, and these pages hash to the value bound in the seal.
4. Each run matches the digest the server received when it ended.
5. A supplied code bundle hashes to the code bundle hash above.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

Appendix

A Full metric series

Run 1: kv_run.sh

Metric	Value	Step	Line	Source
fid_50k	3.64607		60	explicit

Phase	Line	Time	CPU time	Memory	GPU
fid	2	17:51:50	3.5 s	0.58 GB	

B Provenance timeline and activity map

Run 1: kv_run.sh

#	Snapshot	Time	Root	Changed
0	run_start	17:51:48	1dc3d3e61639382b	169
1	phase fid	17:51:50	1dc3d3e61639382b	0
2	run_end	19:55:44	1dc3d3e61639382b	0

Event	Path or command	Hash
read	examples/tabular/README.md	
read	examples/tabular/Tabular_Data_Generation_with_XGBoost_Conditional_Flo...	
read	kv_run.sh	
read	pyproject.toml	
read	requirements.txt	
read	runner-requirements.txt	
read	runner/README.md	
read	runner/configs/callbacks/default.yaml	
read	runner/configs/callbacks/early_stopping.yaml	
read	runner/configs/callbacks/model_checkpoint.yaml	
read	runner/configs/callbacks/model_summary.yaml	
read	runner/configs/callbacks/no_stopping.yaml	
read	runner/configs/callbacks/none.yaml	
read	runner/configs/callbacks/rich_progress_bar.yaml	
read	runner/configs/datamodule/cifar.yaml	
read	runner/configs/datamodule/custom_dist.yaml	
read	runner/configs/datamodule/eb_full.yaml	
read	runner/configs/datamodule/funnel.yaml	
read	runner/configs/datamodule/gaussians.yaml	
read	runner/configs/datamodule/moons.yaml	
read	runner/configs/datamodule/scurve.yaml	
read	runner/configs/datamodule/sklearn.yaml	
read	runner/configs/datamodule/time_dist.yaml	
read	runner/configs/datamodule/torchdyn.yaml	
read	runner/configs/datamodule/tree.yaml	
read	runner/configs/datamodule/twodim.yaml	
read	runner/configs/debug/default.yaml	
read	runner/configs/debug/fdr.yaml	
read	runner/configs/debug/limit.yaml	

Event	Path or command	Hash
read	runner/configs/debug/overfit.yaml	
read	runner/configs/debug/profiler.yaml	
read	runner/configs/eval.yaml	
read	runner/configs/experiment/cfm.yaml	
read	runner/configs/experiment/cnf.yaml	
read	runner/configs/experiment/icnn.yaml	
read	runner/configs/experiment/image_cfm.yaml	
read	runner/configs/experiment/image_fm.yaml	
read	runner/configs/experiment/image_otcfm.yaml	
read	runner/configs/experiment/trajectorynet.yaml	
read	runner/configs/extras/default.yaml	
read	runner/configs/hparams_search/optuna.yaml	
read	runner/configs/hydra/default.yaml	
read	runner/configs/launcher/mila_cluster.yaml	
read	runner/configs/launcher/mila_cpu_cluster.yaml	
read	runner/configs/local/.gitkeep	
read	runner/configs/local/default.yaml	
read	runner/configs/logger/comet.yaml	
read	runner/configs/logger/csv.yaml	
read	runner/configs/logger/many_loggers.yaml	
read	runner/configs/logger/mlflow.yaml	
read	runner/configs/logger/neptune.yaml	
read	runner/configs/logger/tensorboard.yaml	
read	runner/configs/logger/wandb.yaml	
read	runner/configs/model/cfm.yaml	
read	runner/configs/model/cfm_v2.yaml	
read	runner/configs/model/cnf.yaml	
read	runner/configs/model/fm.yaml	
read	runner/configs/model/icnn.yaml	
read	runner/configs/model/image_cfm.yaml	
read	runner/configs/model/otcfm.yaml	
read	runner/configs/model/sbcfm.yaml	
read	runner/configs/model/trajectorynet.yaml	
read	runner/configs/paths/default.yaml	
read	runner/configs/train.yaml	
read	runner/configs/trainer/cpu.yaml	
read	runner/configs/trainer/ddp.yaml	
read	runner/configs/trainer/ddp_sim.yaml	
read	runner/configs/trainer/default.yaml	
read	runner/configs/trainer/gpu.yaml	
read	runner/configs/trainer/mps.yaml	
read	runner/data/.gitkeep	
read	runner/logs/.gitkeep	
read	runner/scripts/schedule.sh	
read	runner/scripts/two-dim-cfm.sh	
read	runner/src/__init__.py	
read	runner/src/datamodules/__init__.py	
read	runner/src/datamodules/cifar10_datamodule.py	
read	runner/src/datamodules/components/__init__.py	
read	runner/src/datamodules/components/base.py	

Event	Path or command	Hash
read	runner/src/datamodules/components/generators2d.py	
read	runner/src/datamodules/components/time_dataset.py	
read	runner/src/datamodules/components/tnet_dataset.py	
read	runner/src/datamodules/components/two_dim.py	
read	runner/src/datamodules/distribution_datamodule.py	
read	runner/src/eval.py	
read	runner/src/models/__init__.py	
read	runner/src/models/cfm_module.py	
read	runner/src/models/components/__init__.py	
read	runner/src/models/components/augmentation.py	
read	runner/src/models/components/base.py	
read	runner/src/models/components/distribution_distances.py	
read	runner/src/models/components/emd.py	
read	runner/src/models/components/evaluation.py	
read	runner/src/models/components/fp16_util.py	
read	runner/src/models/components/hyper_nets.py	
read	runner/src/models/components/icnn_model.py	
read	runner/src/models/components/layers/__init__.py	
read	runner/src/models/components/layers/diffeq_layers/__init__.py	
read	runner/src/models/components/layers/diffeq_layers/basic.py	
read	runner/src/models/components/layers/diffeq_layers/container.py	
read	runner/src/models/components/layers/diffeq_layers/resnet.py	
read	runner/src/models/components/layers/diffeq_layers/wrappers.py	
read	runner/src/models/components/layers/odefunc.py	
read	runner/src/models/components/layers/squeeze.py	
read	runner/src/models/components/logger.py	
read	runner/src/models/components/mlpode.py	
read	runner/src/models/components/mmd.py	
read	runner/src/models/components/nn.py	
read	runner/src/models/components/optimal_transport.py	
read	runner/src/models/components/plotting.py	
read	runner/src/models/components/regularizers.py	
read	runner/src/models/components/schedule.py	
read	runner/src/models/components/simple_dense_net.py	
read	runner/src/models/components/simple_mlp.py	
read	runner/src/models/components/sinkhorn_knopp_unbalanced.py	
read	runner/src/models/components/solver.py	
read	runner/src/models/components/unet.py	
read	runner/src/models/components/utils.py	
read	runner/src/models/icnn_module.py	
read	runner/src/models/runner.py	
read	runner/src/models/utils.py	
read	runner/src/train.py	
read	runner/src/utils/__init__.py	
read	runner/src/utils/pylogger.py	
read	runner/src/utils/rich_utils.py	
read	runner/src/utils/utils.py	
read	runner/tests/__init__.py	
read	runner/tests/conftest.py	
read	runner/tests/helpers/__init__.py	

Event	Path or command	Hash
read	runner/tests/helpers/package_available.py	
read	runner/tests/helpers/run_if.py	
read	runner/tests/helpers/run_sh_command.py	
read	runner/tests/test_configs.py	
read	runner/tests/test_datamodule.py	
read	runner/tests/test_eval.py	
read	runner/tests/test_sweeps.py	
read	runner/tests/test_train.py	
read	setup.py	
read	tests/test_conditional_flow_matcher.py	
read	tests/test_models.py	
read	tests/test_optimal_transport.py	
read	tests/test_time_t.py	
read	torchcfm/__init__.py	
read	torchcfm/conditional_flow_matching.py	
read	torchcfm/models/__init__.py	
read	torchcfm/models/models.py	
read	torchcfm/models/unet/__init__.py	
read	torchcfm/models/unet/fp16_util.py	
read	torchcfm/models/unet/logger.py	
read	torchcfm/models/unet/nn.py	
read	torchcfm/models/unet/unet.py	
read	torchcfm/optimal_transport.py	
read	torchcfm/utils.py	
read	torchcfm/version.py	
read	.gitignore	
read	examples/images/cifar10/compute_fid.py	
read	.github/dependabot.yml	
read	.github/workflows/code-quality.yml	
read	.github/workflows/python-publish.yml	
read	.github/workflows/test.yml	
read	.github/workflows/test_runner.yml	
read	.pre-commit-config.yml	
read	examples/2D_tutorials/Flow_matching_tutorial.ipynb	
read	examples/2D_tutorials/Maximum_likelihood_CNF_tutorial.ipynb	
read	examples/2D_tutorials/SF2M_tutorial.ipynb	
read	examples/2D_tutorials/The_unreasonable_performance_of_minibatch_OT.ip...	
read	examples/2D_tutorials/model-comparison-plotting.ipynb	
read	examples/2D_tutorials/preprocessing/1.0-embryoid_body_data_to_h5ad.ipynb	
read	examples/2D_tutorials/preprocessing/1.1-eb-data-1d-phate.ipynb	
read	examples/2D_tutorials/preprocessing/2.1-split-cite-data.ipynb	
read	examples/2D_tutorials/tutorial_training_8_gaussians_to_moons.ipynb	
read	examples/images/cifar10/train_cifar10.py	
read	examples/images/cifar10/train_cifar10_ddp.py	
read	examples/images/cifar10/utils_cifar.py	
read	examples/images/conditional_mnist.ipynb	
read	examples/images/mnist_example.ipynb	
read	examples/single_cell/single-cell_example.ipynb	
read	.github/PULL_REQUEST_TEMPLATE.md	
read	CODE_OF_CONDUCT.md	

Event	Path or command	Hash
read	LICENSE	
read	README.md	
read	assets/169_generated_samples_otcfm.gif	
read	assets/169_generated_samples_otcfm.png	
read	assets/8gaussians-to-moons.gif	
read	assets/DF_logo.png	
read	assets/DF_logo_dark.png	
read	assets/gaussian-to-moons.gif	
read	examples/2D_tutorials/preprocessing/README.md	
read	examples/images/cifar10/README.md	
process	python -u compute_fid.py --model otcfm --step 400000 --integration_method ...	
process	tee /var/tmp/kv-repro/017-atong01-conditional-flow-matching/fid.out	

C Run history

Invocation 1, OK, 2 hours 3 minutes, 61 output lines, started 2026-10-07T17:51:48

```
metric_hash 8f39b67538c1758d53c25e3f8c6674e76ba6467d1bf8e0a1e08727e021ec2b04
run_digest 4cc6198192f5a1d74ba112f7f176d87204a62e0d9bc93747728e9ceb4dbb6b3d
anchored at run end
```

D Raw seal fields and hashes

Seal

```
data_hash 45245b62b22d14bef244aff320dcb53ce6466f333abdaa5ae41a55bd7e184562
signed_message_hash 2492833625de982ada7ea65427392beea4269a9e3765c5ef0ec37a8a1f0f677b
nonce e591808d7b3350d1b216d5d22e6ec451
signed_at 2026-10-07T19:55:47.551320+00:00
sealed_at 2026-10-07T19:55:47.612657794Z
checkout_bundle 8d2399cb0fe250bae6ec09526af5f450e2875a02ce648669f0657b371dbe60f1
```

Signature (base64)

```
XLkG5jCGi8wm1H4C0/IAMFe66s+17RXbF4m0c9uU17ED2X6CXGafVjvgN9upbpy2Drn9HCK+5qAvqqAsGTGLkpJ
pUm8mCSVD5IJAVrbGtHJ/FwXuCUwdG1li3Z5F1AnYQifP6gLFJHFbrnfDcgHy8u3/Is9hhlpE8eQYhdeVQfk6TDq
Dw5ifvsBzQSF4FRDqCMBNs30vFPQACm7zQjSEPLPQ/DzSClnnHpP1PrKJczbAT+mxeSUD6gE5X0byidUfKHSYCUy
z64pyedGGTJ/WpFSdwf3j6v0oYDTZCdZCHibf09tONGj1IrcEwoVx2KILkhrCFcwUY3zUnZGx20P+mNgsGrYpwc
m+zLM3gn/CqlcKi8+cw8PQ8o4MuYT+A/51JG/U6Cl1GKZOTfbIwyUVWkuAGX9J5Lkw15PElxEsww4aG7b9xtQ5S
BtH7v05myXOPHCxQJu0lxa0uEekilGDPLuuf7jA7/bVm/GQBIC07ka0qzXXxTLdp9WfiGhRSIttisUBxdA/soKAD
IzFSh4KQf8YuiZHQEvV7L1mSBfeIkWMXV2dA2kAtsQigh3jyLdFvi2IXrqFrFp7s23za5xltbgtbrg4NZVuXZC6q
UWGgcR19kBP2WEvkAZj/yJZZL620DuwzcTd5pNFNPFMPDNbWNbcxBb9QMTpz6pgrGSM=
```

Public key

```
-----BEGIN PUBLIC KEY-----
MIICIjANBgkqhkiG9w0BAQEFAAOCAg8AMIICCgKCAgEAL3YyMgeShx8IhTC7jb
sM0w2BnRXgsBn3+SMXKjtwHATqdekmlPlcjayJxD5fcPrOPPCqp/LoUFz5CUZzRi
6+8FTv70dHuIz5A4LAt4vcgwP+iWscPc3HzpmjpsKbTm0emqgp6RB6GNNPpxYa5X
7qU95EfEisX/GfoS9ToxtqPBwCcVwSPBX8efec4TC58m96Fsp4mC8KICeAdBibHN
av0JjdTJ//CM0WZSIbdzE6qvEbNfCqDB5tNU7oHfYvBeNLrvPU9TAqdbtozdHq+D
AhfTRuJNgWekjwD5lb0Vhi4qlkweSOEMTKT3ye9IL5xEaJn4bku4eKqes7VSZt3o
XbUcmr2H5mKcKSvt6bXmX6ajAmgi900Cm9QJlaQkJM6z2iITnyvDMgueJtFxQpC9
HNBPgPZ242NH8JgE6CNF7ToL9syasS67u7UMgIcNgeNHncy9bLgPu5annzP9gN2l
6HEMldkXNdKQRb1rDwyEJf3N/50YHnXXPmuf5sb8R/pxWNqtF04R5UGVV92qd2Mi
1GN7LE3mbQRbPsLL/JApDXMNzNikaFfBsQK7iKxr7vIaTeZckFZWfThcYNT7Hasg
+eE4bf/olCHBCf1VIDckDxohehdZ0WRRcgM9iK3I/0zZdaBsgdc8WuJRmb8LPP9v
8Jap/1PwjNcNku5cdw3gHLcCAwEAAQ==
-----END PUBLIC KEY-----
```

Run 1 outputs

```
stdout  ffceb223457fba99f62380aa2e84852101d24ce124ac30655c331c8b59417c7e
stderr  e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
metrics 8f39b67538c1758d53c25e3f8c6674e76ba6467d1bf8e0a1e08727e021ec2b04
digest  4cc6198192f5a1d74ba112f7f176d87204a62e0d9bc93747728e9ceb4dbb6b3d
```

kv_run.sh (unchanged)

```
pre  8f379b6d9b00ce859736d26ebe21cf7a54702fee0c9563128ceeae4ba5608784
post 8f379b6d9b00ce859736d26ebe21cf7a54702fee0c9563128ceeae4ba5608784
```

E BibTeX for [1]

```
@misc{keita2026computerscienceconferencesrequire,
  title={Computer Science Conferences Should Require Nonrepudiable Experimental Results},
  author={Mamadou K. Keita and Christopher Homan},
  year={2026},
  eprint={2605.08586},
  archivePrefix={arXiv},
  primaryClass={cs.CR},
  url={https://arxiv.org/abs/2605.08586},
}
```